Birla Institute of Technology & Science, Pilani
Pilani Campus

ACADEMIC— UNDERGRADUATE STUDIES DIVISION

SECOND SEMESTER, 2025–2026
Course Handout Part II

Date: 15th September 2025

In addition to Part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No	: CS GXYZ
Course Title	: Agentic AI
Instructor-in-Charge	: Dr. Tejasvi Alladi
Instructors	: Dr. Dhruv Kumar

1. Course Description:

Introduction to Transformers; advanced architectures beyond Transformers (state space models/Mamba, mixture-of-experts, diffusion-based language models); Parameter Efficient Fine-Tuning (PEFT) Basics (LoRA), advanced PEFT (QLoRA, DoRA, AdaLoRA, ReLoRA, quantization-aware adapters); inference and decoding (top-k, top-p, beam, contrastive; vLLM, KV-cache, PagedAttention, AWQ); Model quantization, pruning, distillation, on-device edge optimization; test-time compute (self-consistency, Tree, Graph-of-Thought); post-training & alignment (Reinforcement Learning from Human Feedback (RLHF) pipeline, PPO, REINFORCE, GRPO; direct alignment: DPO, SimPO, KTO, IPO; rejection sampling, instruction tuning, distillation, tool & function calling, Model Context Protocol (MCP)); Retrieval Augmented Generation (RAG) Basics, Advanced RAG (IR foundations, HyDE, RAPTOR, CRAG, DSPy); Agent Security (threat models, attack mechanisms and defense strategies, red teaming, blue teaming);evolutionary/genetic methods (prompt & agent evolution); multimodal LLMs & agents (Flamingo, BLIP-2, LLaVA; MM-ReAct, ViperGPT).

2. Scope & Objectives of the Course:

- Compare and justify LLM/agent architecture choices for long-context, latency, and reasoning.
- Implement and evaluate PEFT methods under memory/compute constraints.
- Design decoding and test-time compute strategies within fixed latency/cost budgets.
- Build and align models/agents using RLHF and direct-alignment methods; diagnose reward hacking and length bias.
- Construct robust, programmable RAG agent pipelines with provenance and corrective feedback.
- Analyze and mitigate AI agent security threats, conduct red & blue teaming.
- Apply evolutionary search to improve prompts, tools, MCP usage and agent policies.
- Fine-tune and evaluate multimodal LLMs/agents; integrate perception with tool-use plans safely.

3. Course Learning Outcomes:

At the end of the course, the students will be able to:

- Explain and compare advanced LLM/agent architectures (Transformers, SSMs, Mamba, MoE, diffusion etc.).
- Apply parameter–efficient fine-tuning (LoRA, QLoRA, DoRA, AdaLoRA, ReLoRA) under memory and compute constraints.
- Configure decoding and serving (top-k, top-p, beam; vLLM, KV-cache, PagedAttention) to meet latency and cost targets.
- Implement test-time compute strategies (self-consistency, Tree/Graph-of-Thought) and evaluate accuracy–cost trade-offs.
- Build post-training model/agent pipelines (SFT $\rightarrow$ reward modeling $\rightarrow$ RLHF or direct alignment) and mitigate reward hacking/length bias.
- Use direct-alignment methods (DPO, SimPO, KTO, IPO) and rejection sampling to improve helpfulness and safety.
- Design robust RAG agentic systems (IR foundations, HyDE, RAPTOR, CRAG, DSPy) with provenance and corrective feedback.
- Evaluate and defend AI agents against adversarial threats (prompt injection, data poisoning, tool misuse); perform red & blue teaming.
- Employ evolutionary methods to optimize prompts, tools, MCP usage and agent policies against defined fitness metrics.
- Fine-tune and assess multimodal LLMs/agents (e.g., LLaVA, MM-ReAct) with appropriate safety guardrails.

4. Textbooks:

TB1. *Reinforcement Learning from Human Feedback*, Nathan Lambert, First Edition, Online, 2025 [Online Link]

TB2. *Introduction to Large Language Models*, Tanmoy Chakraborty, First Edition, Wiley, 2024 [Online Link]

Reference Textbooks:

RB1. *Generative Deep Learning (2nd Ed.)*, David Foster, O'Reilly, 2023 [Online Link]

RB2. *Reinforcement Learning: An Introduction (2nd Ed.)*, Richard S. Sutton, Andrew G. Barto, MIT Press, 2018 [Online Link]

RB3. Research Papers published in Top CS venues (ICLR, ICML, NeurIPS, ACL, EMNLP, NAACL etc.)

5. Course Plan:

Week no.	Module No.	Lecture No.	Reference	Learning Outcomes
1.	Transformers, Self-Attention, Multi-Head Attention	1–3	TB2	Explain attention and multi-head attention; Train a transformer-based language model.
2.	Beyond Transformers I: State Space Models, Mixture of Experts & Diffusion-based Language Models	4–6	RB1, RB3	Explain limits of attention, derive SSM/Mamba update rules, compare dense vs sparse-expert scaling, analyze routing/latency trade-offs. Contrast Auto-regressive vs diffusion-based model training/sampling for text.
3.	Advanced PEFT and Model Compression: PEFT Basics, LoRA, QLoRA, & DoRA, AdaLoRA and Q-aware LoRA	7–9	TB2, RB3	Implement PEFT and model compression (pruning, quantization, distillation) to minimize training memory and compute, quantization-aware fine-tuning, pruning, and distillation.
4.	Inference and Edge Optimization & Test-Time Compute	10–12	TB2, RB3	Choose/justify decoding (top-k, top-p, beam, typical), design serving with KV-cache & paged attention, quantize with minimal loss, apply TTC (self-consistency/ToT) under latency budgets.
5.	Post-Training I: RLHF Overview, Data, Rewards	13–15	TB1, RB2, RB3	Map SFT→RM→RL, DPO pipeline, design preference datasets, diagnose reward overoptimization & calibration.
6.	Post-Training II: PPO/REINFORCE, GRPO, Rejection Sampling	16–17	TB1, RB2, RB3	Implement policy-gradient variants for LMs, compare stability/sample efficiency, use rejection sampling baselines for reasoning.
7.	Post-Training III: Direct Alignment (DPO family), Instruction Tuning, Tools	19–21	TB1, RB2, RB3	Derive DPO objective, train reference-free variants, apply alternative utilities, control length/verbosity via regularizers, build instruction mixtures, distill synthetic data, align tool-use via correctness rewards, Model Context Protocol (MCP)
8.	Interpretability and Explainability	22–24	RB3	Mechanistic interpretability (circuits, activation patching, causal tracing), Representation probing and layer analysis.
9.	Advanced RAG I: RAG Basics, IR Foundations to HyDE/RAPTOR	25–27	TB2, RB3	Build a RAG Pipeline, Compare BM25 vs dense/ANN, engineer query-side hypotheses (HyDE), construct hierarchical chunk trees (RAPTOR) for long documents.

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Week no.	Module No.	Lecture No.	Reference	Learning Outcomes
10.	Advanced RAG II: Robustness, CRAG, DSPy	28–30	TB2, RB3	Add corrective feedback loops, track provenance & trust signals, program retrieval chains with declarative optimization.
11.	AI Agent Security	31–34	TB2, RB3	Agent threat models, red teaming and adversarial testing, defense and guardrails, policy-based and interpretability-based defenses; MCP security.
12.	Evolutionary & Genetic Methods with LLMs	34–36	RB3	Design mutation/selection for prompts/tools, set fitness (win-rate, reward, safety), evolve curricula and toolchains for tasks, Self-evolving LLMs.
13.	Multimodal LLMs I: Architectures	37–39	TB2, RB1, RB3	Contrast perceiver-resampler, Q-former, and adapter designs; map VLM/VLA training data & capabilities; identify hallucination failure modes.
14.	Multimodal LLMs II: Post-Training & Agents	40–42	TB2, RB1, RB3	Conduct visual instruction tuning, integrate perception with tool plans, discuss safety alignment for multimodal agents.

6. Evaluation Scheme:

Component	Wt (%)	Duration	Date & Time	Comments
Mid-semester Examination	25	90 mins	Time Table	Partly Open Book
Comprehensive Examination	35	180 mins.	Time Table	Partly Open Book
Labs	10	2 hours per week	Time Table	Will include reviewing and reproducing case studies from OpenAI, Anthropic, Google etc.
Research Project	30	-	-	Well-defined evaluation criteria along with multiple deadlines throughout the semester. Also, includes paper presentation and review.

NC Criterion: Any student receiving less than 10% of the maximum course marks will be awarded a NC grade.

7. Attendance Policy:

Attendance in the lectures is highly encouraged. But there are no marks associated with attendance in the lectures.

8. Office Consultation Hour:

Will be announced in the class. Also, feel free to email me to seek a prior appointment.

9. Make-up Policy:

Make-up tests will be given only to genuine cases.

Signature
Instructor-in-charge
Course number and Title